Original Article

Container shipping route design incorporating the costs of shipping, inland/feeder transport, inventory and CO_2 emission

Nguyen Khoi Tran[a,b], Hans-Dietrich Haasis[a,b] and Tobias Buer[a,c]

[a]Faculty of Business Studies & Economics, University of Bremen, Wilhelm-Herbst-Str. 12, Bremen 28359, Germany.
E-mails: khoi.tran@uni-bremen.de; haasis@uni-bremen.de; tobias.buer@uni-bremen.de
[b]Chair in Maritime Business and Logistics.
[c]Computational Logistics, Cooperative Junior Research Group of University of Bremen and ISL - Institute of Shipping Economics and Logistics, Germany.

Abstract As container shipping networks have become important components in global supply chains, route design should take both maritime and inland factors into consideration. In this article, a model is proposed to optimise container flows between two continents via an end-to-end service. The model is concerned not only with the design of an optimal shipping route but also with inland connections between hinterlands and ports. The objective is to minimise total costs, consisting of ship costs, port costs, inland/feeder transport costs, inventory costs and CO_2 costs. The model is applied to the actual trade between Europe and the United States. Computational outcomes show that ship costs and port costs (port dues and terminal handling charges) represent less than one third of total costs. Therefore, the *maritime network* is only a part of a bigger system and piecemeal optimisation may not guarantee the optimisation of the whole network. Inland/feeder transport costs contribute the most to total costs, and they are influenced significantly by port choice. Although the use of a greater number of ports results in longer distances and higher shipping costs, this benefits in terms of lower distribution costs between hinterlands and ports. Inventory costs play a considerable part in total costs and they increase as vessel capacity goes up. In other words, these costs present a barrier to the introduction of bigger vessels. Optimal size is obviously a trade-offs between inventory and shipping costs.

Maritime Economics & Logistics advance online publication, 31 March 2016; doi:10.1057/mel.2016.11

© 2016 Macmillan Publishers Ltd. 1479-2931

Keywords: liner shipping; global supply chains; route design; mega vessels; inventory costs; environmental costs

Introduction

Ports have developed beyond their traditional boundaries. They are no longer interfaces between shores and ships, or the places where cargo transportation starts or finishes, but have become nodes in seamless cargo flows from manufacturers to consumers. This tendency has been clearly illustrated in port evolution models (see for example UNCTAD, 1992; Robinson, 2002, 2006; Carbone and De Martino, 2003; Notteboom and Rodrique, 2008; Pettit and Beresford, 2009).

As ports are being embedded in global supply chains, shipping network design should be placed in a broader scope, not only within maritime transportation but also in the interaction with inland transportation. Maritime factors regarding ship and port operations may not be enough to determine an optimal route. Other logistics factors such as inland transportation, value of cargo, service frequency and even environmental effects should be taken into consideration as well.

Numerous researches have tackled network optimisation in container shipping. Most of them have focused on the maritime leg. This article approaches the optimisation issue in a broader scope. Key decisive parameters in our model consist not only of conventional ship and port costs but also inland/feeder transport costs, inventory costs and CO_2 costs.

The remainder of this article is organised as follows. The next section provides a review of previous studies. The optimisation model is described in the following section. Afterwards, the data and the operational parameters are introduced. The penultimate section presents experiment results and some in-depth analyses. The final section consists of the conclusions.

Literature Review

Shipping network design is an important theme in the research on container shipping. One key branch consists of the empirical research aimed to discover network characteristics, as well as the strategies of carriers in organising their services (see for example Robinson, 1998; Fremont, 2007; Ferrari *et al*, 2008; Cullinane and Wang, 2012; Ducruet, 2013; Seoane *et al*, 2013; Tran and Haasis, 2014). Another branch of research is to propose mathematical models to optimise network design. A comprehensive list of optimisation papers can be found in review works (Christiansen *et al*, 2004, 2007, 2013; Kjeldsen, 2011; Meng *et al*, 2013; Tran and Haasis, 2015b).

Route design is related to choosing appropriate port rotations, and the scheduling of shipping routes. Wijnolst *et al* (2000) select calling hubs, on the route of Malacca-max ships, between Far East and Europe, based on the model of the centre of gravity. Aversa *et al* (2005) used a p-hub median problem to choose suitable hub ports in South America. Baird (2006) develops the model of 'deviation cost' to evaluate a new transhipment hub in Northern Europe. Imai *et al* (2006) study the viability of mega vessels on hub-and-spoke (H&S) configurations and multiple-port-calls (MPC). Gelareh *et al* (2010), Gelareh and Pisinger (2011) and Gelareh *et al* (2013) establish models to optimise H&S systems. Meng and Wang (2011b) incorporate the problems of route selection, fleet assignment and container routing in the combined H&S and MPC system. Polat *et al* (2014) address the design of feeder routes in the Black Sea region.

Shintani *et al* (2007) propose a two-stage model to determine an optimal sequence of visited ports on a shipping route. Verni and Grigentin (2009) analyse the feasibility of container shipping passage via the Arctic Sea. Cariou and Cheaitou (2014) optimise liner shipping services between Northern Europe and East Coast South America, taking cascading effects into account. Tran (2011) considers inland origin/destination of shipments in designing a service to carry containers on Trans-Atlantic trade without being concerned with the impact of environmental factors and service frequency on route design. In addition, some cost models regarding shipping operations seem to be unreasonable. Liu *et al* (2014) consider the shipping network as part of an intermodal network and propose a model to select suitable shipping routes. Mulder and Dekker (2014) develop the concept of *port cluster* including a group of regional ports to reduce the size of network design problems. Yang *et al* (2012) optimise route design by paying attention to the interaction between shipping services and port demand.

Qi and Song (2012) take account of port uncertainty and delay cost of vessels in optimising sailing speed. Meng and Wang (2011a) optimised sailing speed and service frequency on an Asia–Europe shipping route. In addition to sailing speed, the model of Wang and Meng (2012c) selects the arrival schedule of shipping routes in the context of port and sea uncertainty. Brouer *et al* (2012) address solutions to maintain vessel schedule in case of disruptions. Wang *et al* (2014) determine arrival and departure times of a Trans-Atlantic route under the limitation of port time windows.

Fleet management is concerned with employing appropriate ship size, assigning ships to routes, investing in new capacity, chartering needed capacity and leasing out surplus one. Mourao *et al* (2002) focus on assigning fleet capacity on a H&S network. Also working on a H&S network, Zacharioudakis *et al* (2011) create a non-linear programming model to allocate fleet tonnage and select appropriate speeds. Gelareh and Meng (2010) deal with assignment and lay-ups, and charter-in and out decisions. Meng and Wang (2011c) study ship assignment

in multiple periods. Meng *et al* (2015) address fleet planning not only in multiple periods but also in cases of stochastic transport demand.

In container shipping, carriers must work with a large number of customers who have different requirements regarding the origin/destination of goods, type of containers, and transit time constraints. They must take account of both laden and empty containers. Song and Dong (2012) optimise the movement of loaded and empty boxes on multiple routes to minimise handling cost, lost-sales cost, waiting cost at transhipment ports, container cost and empty container repositioning cost. Bell *et al* (2011, 2013) build models to optimise container flows taking transhipment operation between routes into consideration. Their former model aims to minimise total sailing time and container dwell time in ports, whereas the latter aims to minimise handling cost, rental cost and inventory cost of containers. Such models have been extended by Wang *et al* (2015) by considering elastic demand, as well as trying to maximise the profit of operators. Wang and Tang (2010) and Lofstedt *et al* (2010) deal with optimal transportation of laden and empty containers in multiple periods and multiple shipping routes. Wang *et al* (2014) solve the joint problem of ship scheduling and cargo routing to satisfy delivery time of goods. Wang (2014) propose a hybrid-link-based model to reduce the number of decision variables in container routing problems.

Reinhardt *et al* (2007) formulate a mixed integer programming model to solve network design and fleet deployment problems with the consideration of transhipment between routes. Alvarez (2009) proposes a two-tier model to deal with network design. Agarwal and Ergun (2008) take account of weekly frequency and weekday constraints in their network problem. Their model is further developed (Agarwal and Ergun, 2010) to solve the network problem of shipping alliances. The models of Reinhardt and Pisinger (2012) and Plum *et al* (2014) are based on butterfly routes, that is, when a port can be visited two times or more. Wang (2013) examines the impact of slot-purchasing, integer number of containers, container type, empty container repositioning and ship repositioning on network design and fleet deployment. Song and Dong (2013) incorporate route structure, ship deployment and empty container repositioning in their optimisation model for long-haul liner service design. Wang and Meng (2014) tackle the network design problem by considering delivery deadlines of cargo.

Network design has attracted much attention, especially in recent years. This topic has been approached from various directions and methodologies. However, network problems have been mostly solved for the maritime leg with only little consideration to the evolving role of ports on the global supply chain. This article approaches the gap in the literature. Conventional models are often based on cargo flows between ports, our model deals further with the flows between hinterlands. In addition, application on real data facilitates the estimation of components of supply chain costs, as well as evaluates the impact of different operational factors on network optimisation.

Optimisation Model

Description

Our model deals with container trade between two regions A and B. Each region is divided into different hinterlands. Container flows between hinterlands have been classified with information regarding the number of TEUs and cargo values in a specific period of time. There are some candidate ports that can be visited by mainline ships. Goods are assumed to be carried via an end-to-end shipping service with a pre-determined ship size.[1]

The aim of our model is to design not only an optimal shipping service but also optimal inland connections between hinterlands and ports. The objective is to minimise total costs consisting of ship costs (capital cost, operational cost and bunker cost); port costs (port dues and terminal handling charges); inventory costs (time cost of cargo, container cost and safety stock); inland/feeder transportation costs between original/final points and loading/unloading ports; and CO_2 costs (generated by ship and port operations, and inland /feeder transportation). Three questions are considered in the model: (i) Which ports should be served along the shipping route? (ii) What should be the sequence of port calls? (iii) What are the loading and unloading ports of a shipment? The first two questions are related to the organisation of a shipping route, whereas the third one is related to the inland connections. An important characteristic of our model is to consider not only the costs of shipping lines but also those of shippers. An optimal service is actually a balance between their benefits Figure 1.

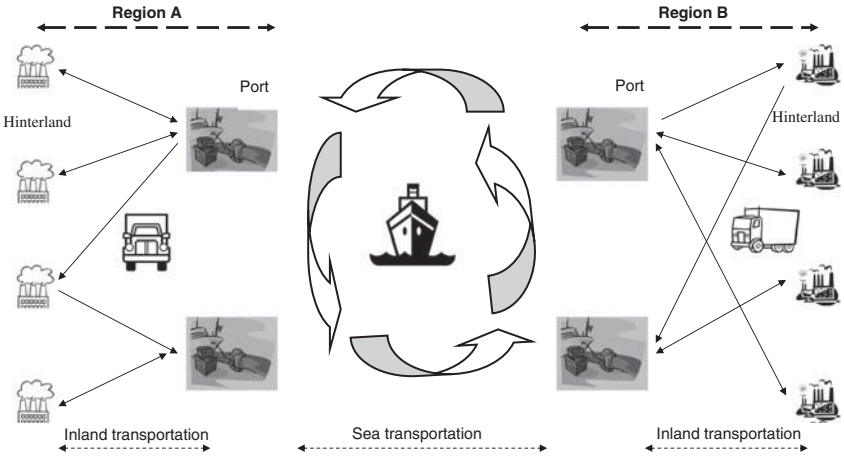


Figure 1: Transportation network.

Mathematical model

Objective: To minimise total costs of ship costs, port costs, inland/feeder transport costs, inventory costs and CO_2 costs.

$$\text{minimise } TC = TSC + TPC + TILC + TIC + TCC$$

$$= \sum_{\alpha,\beta \in HUB} \frac{sd_{\alpha,\beta} * lk_{\alpha,\beta} * ssc}{v} + \sum_{\alpha \in HUB} pt_{\alpha} * spc + \sum_{\alpha \in HUB} due_{\alpha}$$

$$+ \sum_{\alpha \in HUB} thc_{\alpha} * \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{box_{i,j}\left(ld_{i,j,\alpha} + uld_{i,j,\alpha}\right)}{voy}$$

$$+ \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{value_{i,j}}{voy} * time_{i,j} * inv$$

$$+ \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}}{voy} * time_{i,j} * cc + \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{value_{i,j}}{TP} * \frac{freq}{2} * inv$$

$$+ \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(ld_{i,j,\alpha} * ic_{i,\alpha} + uld_{i,j,\alpha} * ic_{j,\alpha}\right)}{voy}$$

$$+ \sum_{\alpha,\beta \in HUB} \frac{sd_{\alpha,\beta} * lk_{\alpha,\beta}}{v} * sCO2 + \sum_{\alpha \in HUB} pt_{\alpha} * pCO2$$

$$+ \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(ld_{i,j,\alpha} + uld_{i,j,\alpha}\right)}{voy} * hCO2_{\alpha}$$

$$+ \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(iCO2_{i,\alpha} * ld_{i,j,\alpha} + iCO2_{j,\alpha} * uld_{i,j,\alpha}\right)}{voy}$$

By changing *binary* variables:

$$ld_{i,j,\alpha} \qquad i = \overline{1, N+M};\ j = \overline{1, N+M};\ \alpha = \overline{1, K+T}$$

$$uld_{i,j,\alpha} \qquad i = \overline{1, N+M};\ j = \overline{1, N+M};\ \alpha = \overline{1, K+T}$$

$$rt_{\alpha} \qquad \alpha = \overline{1, K+T}$$

$$lk_{\alpha,\beta} \qquad \alpha = \overline{1, K+T};\ \beta = \overline{1, K+T}$$

 © 2016 Macmillan Publishers Ltd. 1479-2931

Constraints:

$$\forall \alpha \in HUB \sum_{\beta \in HUB} lk_{\alpha,\beta} = 1 \tag{1}$$

/* *each selected port has exactly one previous port and one subsequent port on the voyage.*

$$\forall \alpha \in HUBlk_{\alpha,\beta} = 0 \tag{2}$$

/**no self- connection from a port to itself.*

$$\forall \alpha \notin HUB \sum_{\beta=1}^{K+T} lk_{\alpha,\beta} = 0 \tag{3}$$

/* *non-selected port has no one-way sea connection with other ports.*

$$\forall \alpha \notin HUB \sum_{\beta=1}^{K+T} lk_{\beta,\alpha} = 0 \tag{4}$$

/* *no port has one-way sea connection with non-selected ports.*

$$\begin{aligned} \sum_{\alpha \in HUBA} \sum_{\beta \in HUBA} lk_{\alpha,\beta} &= \sum_{\alpha \in HUBA} rt_\alpha - 1 \\ \sum_{\alpha \in HUBB} \sum_{\beta \in HUBB} lk_{\alpha,\beta} &= \sum_{\alpha \in HUBB} rt_\alpha - 1 \end{aligned} \tag{5}$$

/* *the ship calls at all selected ports in a region, then travels to another region*

$$\forall (i,j) \in OD \sum_{\alpha \in HUB} ld_{i,j,\alpha} * (1 - |lh_i - lp_\alpha|) = 1 \tag{6}$$

/* *cargo from i to j is loaded by exactly one port, loading port α is in the same region with hinterland i.*

$$\forall (i,\ j) \in OD \sum_{\alpha \in HUB} uld_{i,j,\alpha} * \left(1 - \left|lh_j - lp_\alpha\right|\right) = 1 \tag{7}$$

/* *cargo from i to j is unloaded by exactly one port, unloading port α is in the same region with hinterland j.*

$$\forall (i,j) \in OD \sum_{\alpha \notin HUB} ld_{i,j,\alpha} = 0 \qquad \sum_{\alpha \notin HUB} uld_{i,j,\alpha} = 0 \tag{8}$$

/* *non-selected ports are not loading and unloading ports for any shipment.*

$$\forall (i,j) \notin OD \sum_{\alpha=1}^{K+T} ld_{i,j,\alpha} = 0 \qquad \sum_{\alpha=1}^{K+T} uld_{i,j,\alpha} = 0 \tag{9}$$

/* *there is no loading and unloading port for any pair (i, j) with no cargo transportation demand.*

Model variables

Decision variables

$rt_\alpha = 1$ Port α is selected on the route; or else $rt_\alpha = 0$

$lk_{\alpha,\beta} = 1$ Port β is the subsequent port of α on the voyage; or else $lk_{\alpha,\beta} = 0$

$ld_{i,j,\alpha} = 1$ Containers from i to j are loaded at port α; or else $ld_{i,j,\alpha} = 0$

$uld_{i,j,\alpha} = 1$ Containers from i to j are unloaded by port α; or else $uld_{i,j,\alpha} = 0$

Input variables

They are the input for the computation and are extracted from actual data (See section *Data description* for details).

N Number of hinterlands in region A

M Number of hinterlands in region B

Hinterlands in A are numbered from 1 to N, hinterlands in B from $N+1$ to $N+M$

$$lh_i = 0 \; if \; 1 \leqslant i \leqslant N; \; or \; else \; lh_i = 1$$

K The number of candidate ports in region A

T The number of candidate ports in region B

Ports in A are numbered from 1 to K, ports in B from $K+1$ to $K+T$

$$lp_\alpha = 0 \; if \; 1 \leqslant \alpha \leqslant K; \; or \; else \; lp_\alpha = 1$$

$teu_{i,j}$ The number of TEUs from hinterland i to hinterland j (TEU). It represents the amount of cargo to be transported between the two hinterlands.

$box_{i,j}$ The number of boxes (containers) deployed to carry cargo from i to j (container). Boxes have different sizes, the two basic ones being 20 foot (1 TEU) and 40 foot (2 TEUs). Whereas other works are often based on the number of TEUs, to calculate the number of handling operations between ship and shore, our research is based on the number of boxes.

$value_{i,j}$ Total cargo value from hinterland i to hinterland j (USD)

TP Time period of transportation demand (days). Our model deals with container flows between the United States and Europe in a month.

size Ship capacity (TEU)

uti Ship utilisation on the main leg (per cent)

voy The number of round voyages in the time period of transportation demand (*TP*). This is calculated by the ratio between the total shipping volume on the main direction (with higher traffic) and the volume a ship can carry per trip.

$$voy = \max\left\{\frac{\sum_{i=1}^{N}\sum_{j=N+1}^{N+M} teu_{i,j}}{uti^{*}size};\ \frac{\sum_{i=N+1}^{N+M}\sum_{j=1}^{N} teu_{i,j}}{uti^{*}size}\right\}$$

freq The interval between two consecutive services (day) $freq = (TP)/(voy)$

inv Inventory costs as percentage of cargo value (per cent/day)

v Ship speed (miles/day)

cc Daily costs per TEU (USD)

due_{α} Port dues (berth due, pilotage, towage and so on) per ship call in port α (USD/call)

thc_{α} Terminal handling charge in port α (USD/move). Whereas port dues per ship call are fixed based on a ship's design capacity, terminal handling charges vary based on the number of containers loaded/unloaded at the port.

pro_{α} Handling productivity of gantry cranes to load and unload containers between ship and shore per day in port α (moves/day). This indicator, together with manoeuvring time, determines time in port of a ship.

edt_{α} Dwell time of export cargo in port α before ship operation (day)

idt_{α} Dwell time of import cargo in port α after ship operation (day)

mt_{α} Manoeuvring time per entry/exit in port α (day)

$hCO2_{\alpha}$ CO_2 cost for serving one TEU in port α (USD); it is based on the amount of CO_2 emission per TEU in the port generated by its operation and CO_2 tax.

$ic_{i,\alpha}$ Inland/transhipment cost between hinterland i and port α (USD/TEU)

$it_{i,\alpha}$ Inland/transhipment time between hinterland i and port α (day)

$iCO2_{i,\alpha}$ CO_2 cost between hinterland i and port α generated by inland/transhipment operation (USD/TEU)

$sd_{\alpha,\beta}$ Nautical distance between port α and port β (mile)

ssc Daily ship cost at sea (USD)

spc Daily ship cost in port (USD)

sCO2 Daily CO_2 cost generated by ship operation at sea (USD)

pCO2 Daily CO_2 cost generated by ship operation in port (USD)

Intermediate variables:

HUB Set of selected ports

HUBA Set of selected ports in region A

HUBB Set of selected ports in region B

OD Set of cargo flows. $OD = \{(i,j)|1 \leqslant i,j \leqslant N+M; teu_{i,j} > 0\}$

$r_{\alpha,\beta,\gamma}=1$ The ship passes through port γ on the voyage from α to β; or else $r_{\alpha,\beta,\gamma}=0$

pt_{α} Total time the ship spends in port α. It includes manoeuvring time and unloading and loading time (day)

$$pt_{\alpha}=2{*}mt_{\alpha}+\frac{\sum_{i=1}^{N+M}\sum_{j=1}^{N+M}box_{i,j}{*}ld_{i,j,\alpha}+\sum_{i=1}^{N+M}\sum_{j=1}^{N+M}box_{i,j}{*}uld_{i,j,\alpha}}{pro_{\alpha}{*}voy}$$

$tst_{\alpha,\beta}$ Total sea time from port α to port β (day)

$$tst_{\alpha,\beta}=\sum_{\gamma\in HUB}\sum_{\delta\in HUB}\frac{sd_{\gamma,\delta}{*}lk_{\gamma,\delta}{*}r_{\alpha,\beta,\gamma}{*}r_{\alpha,\beta,\delta}}{v}$$

$tpt_{\alpha,\beta}$ Total port time on the voyage from port α to port β (day)

$$tpt_{\alpha,\beta}=\sum_{\gamma\in HUB}pt_{\gamma}{*}r_{\alpha,\beta,\gamma}$$

$time_{i,j}$ Total transit time from hinterland i to hinterland j (day). It includes (i) transit time from the point of origin to the loading port, (ii) dwell time in the loading port before ship operation, (iii) time from the ship arriving to the loading port until it leaves the unloading port, (iv) dwell time in the unloading port after ship operation and (v) transit time from the unloading port to the final destination.

$$time_{i,j}=\sum_{\alpha\in HUB}it_{i,\alpha}{*}ld_{i,j,\alpha}+\sum_{\alpha\in HUB}edt_{\alpha}{*}ld_{i,j,\alpha}+\sum_{\alpha,\beta\in HUB}\left(st_{\alpha,\beta}+pt_{\alpha,\beta}\right){*}ld_{i,j,\alpha}{*}uld_{i,j,\beta}+\sum_{\beta\in HUB}idt_{\beta}{*}uld_{i,j,\beta}+\sum_{\beta\in HUB}it_{j,\beta}{*}uld_{i,j,\beta}$$

TSC Total ship costs at sea and in port (USD)

$$TSC=\sum_{\alpha,\beta\in HUB}\frac{sd_{\alpha,\beta}{*}lk_{\alpha,\beta}{*}ssc}{v}+\sum_{\alpha\in HUB}pt_{\alpha}{*}spc$$

TPC Total port dues and terminal handling charge (USD)

$$TPC=\sum_{\alpha\in HUB}due_{\alpha}+\sum_{\alpha\in HUB}thc_{\alpha}{*}\sum_{i=1}^{N+M}\sum_{j=1}^{N+M}\frac{box_{i,j}\left(ld_{i,j,\alpha}+uld_{i,j,\alpha}\right)}{voy}$$

TCC Total CO_2 costs (USD). They include CO_2 cost at sea and in port generated by ship operations, CO_2 cost generated by cargo operation in port and

CO_2 cost generated by inland /feeder transportation.

$$TCC = \sum_{\alpha,\beta \in HUB} \frac{sd_{\alpha,\beta} * lk_{\alpha,\beta}}{v} * sCO2 + \sum_{\alpha \in HUB} pt_{\alpha} * pCO2 + \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(ld_{i,j,\alpha} + uld_{i,j,\alpha}\right)}{voy} * hCO2_{\alpha} + \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(iCO2_{i,\alpha} * ld_{i,j,\alpha} + iCO2_{j,\alpha} * uld_{i,j,\alpha}\right)}{voy}$$

TIC Total inventory carrying costs, container cost and safety stock (USD). Inventory carrying costs depend on cargo value and transit time from origin to destination. Container cost is paid for the use of containers. Safety stock regards the prevention of *out of stock* between two consecutive trips and it is determined by service frequency. It is calculated by the formula taken from Jansson and Shneerson(1987) without consideration of storage costs.

$$TIC = \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{value_{i,j}}{voy} * time_{i,j} * inv + \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}}{voy} * time_{i,j} * cc + \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{value_{i,j}}{TP} * \frac{freq}{2} * inv$$

TILC Total inland/feeder transportation costs (USD)

$$TILC = \sum_{\alpha \in HUB} \sum_{i=1}^{N+M} \sum_{j=1}^{N+M} \frac{teu_{i,j}\left(ld_{i,j,\alpha} * ic_{i,\alpha} + uld_{i,j,\alpha} * ic_{j,\alpha}\right)}{voy}$$

Solution algorithm

A brute-force algorithm and a greedy algorithm are used to solve the non-deterministic polynomial problem. The former algorithm generates all possible ship voyages. For each generated voyage, the latter algorithm is applied to allocate loading and unloading ports for each shipment. On the basis of ship voyage and inland transportation flows, the total cost for each option is calculated. The option with minimum total cost is selected as the solution.

There are five major steps in this approach:

Step 1: Port choice

On the basis of the list of candidate ports, all possibilities of port choice are created with $2^K - 1$ options in region A, and $2^T - 1$ options in region B. The total number of options is $(2^K - 1) * (2^T - 1)$. Table 1 illustrates the list of choices when

Table 1: List of port choices

Region A				*Region B*			
	Port choice		*Port choice*		*Port choice*		*Port choice*
1	1	9	2	1	5	9	6
2	1, 2	10	2, 3	2	5, 6	10	6, 7
3	1, 2, 3	11	2, 3, 4	3	5, 6, 7	11	6, 7, 8
4	1, 2, 3, 4	12	2, 4	4	5, 6, 7, 8	12	6, 8
5	1, 2, 4	13	3	5	5, 6, 8	13	7
6	1, 3	14	3, 4	6	5, 7	14	7, 8
7	1, 3, 4	15	4	7	5, 7, 8	15	8
8	1, 4			8	5, 8		

Table 2: A set of ship voyages

1	1→3 →4→5→8→1	7	3→4 →1→5→8→3
2	1→3 →4→8→5→1	8	3→4 →1→8→5→3
3	1→4 →3→5→8→1	9	4→3 →1→5→8→4
4	1→4 →3→8→5→1	10	4→3 →1→8→5→4
5	3→1 →4→5→8→3	11	4→1 →3→5→8→4
6	3→1 →4→8→5→3	12	4→1 →3→8→5→4

each region has 4 candidate ports (15 cases per region). There are 225 (15*15) ways to select ports on the ship's voyage.

Step 2: Route generation

Corresponding to each state in step 1, different sequences of port calls are enumerated by generating all permutations of the selected ports in each region. Totally, there are $\sum_{i=1}^{K}(K!/(K-i)!)^{*}\sum_{j=1}^{T}(T!/(T-i)!)$ potential ship voyages. Table 2 displays possible voyages in case ports 1, 3 and 4 are chosen in region A, ports 5 and 8 in region B.

Step 3: Selection of loading and unloading ports

For each voyage generated in step 2, four corresponding costs are defined (Table 3): ship costs at sea (1), port dues (3) and CO_2 cost at sea (11) and safety stock (8). Other costs are determined when the loading and unloading ports, for all shipments, are allocated. The optimal state of each voyage obviously depends on the choice of loading and discharging ports. Our greedy strategy is not meant to optimise all the undefined costs, but only some of them. The greedy cost of a shipment, corresponding to a pair of loading and unloading ports, is defined as the sum of terminal handling charges (4), inventory costs at sea (5), inland/feeder transport costs (9), CO_2 costs generated by inland/feeder transport (12) and handling operation in port (13). The greedy idea is based on the fact that the greedy cost of each shipment is independent of the choice of loading and

Table 3: Total cost breakdown

	Cost	*Stage*	*Defined*
1	Ship cost	At sea	No
2	—	In port	Yes
3	Port cost	Port dues	Yes
4		THC	No
5	Inventory cost	At sea	No
6	—	In port	No
7	—	Inland/feeder	No
8	—	Safety stock	Yes
9	Inland/feeder transport cost	—	No
10	CO_2 cost (ship)	At sea	Yes
11	—	In port	No
12	CO_2 cost (cargo)	Inland/feeder	No
13	—	In port	No

unloading ports of other shipments. Therefore, instead of solving a large problem for all shipments, we try to find a local optimum for each shipment by choosing the loading and unloading ports with minimum greedy cost. Ship costs (2), inventory costs (6) and CO_2 cost (11) in port are, therefore, out of consideration in this step and can be referred to as model error. In the next steps, these missing costs are still taken into account to calculate the total voyage cost.

The greedy cost of the shipment from i to j with loading port α and unloading port β is:

$$gc_{i,j,\alpha,\beta} = \left(ic_{i,\alpha} + ic_{j,\beta}\right) * \frac{teu_{i,j}}{voy} + \left(iCO2_{i,\alpha} + iCO2_{j,\beta} + hCO2_{\alpha} + hCO2_{\beta}\right) * \frac{teu_{i,j}}{voy} + \left(\frac{value_{i,j}}{voy} * inv + \frac{teu_{i,j}}{voy} * cc\right) * \left(it_{i,\alpha} + edt_{\alpha} + tst_{\alpha,\beta} + idt_{\beta} + it_{j,\beta}\right) + \left(thc_{\alpha} + thc_{\beta}\right) * \frac{box_{i,j}}{voy}$$

Step 4: Cost estimation – On the basis of the ship route and loading and unloading ports for shipments, the total cost of a voyage is calculated.

Step 5: Update – If the total cost is smaller than the recorded result, the new option will be updated.

Data Description

Cargo flow

Raw data were obtained by the Port Import Export Reporting Service (PIERS). They include 108 847 monthly seaborne trade profiles between the United States

and European countries (France, the United Kingdom, The Netherlands, Germany, Belgium, Norway, Sweden and Denmark). Each profile provides the basic information of a shipment: origin, destination, the number of TEUs and containers and value. The origin or destination point of a shipment is a city in Europe and a state in the United States. On the basis of these, 3179 hinterlands are classified in Europe (on the basis of city) and 46 in the United States (on the basis of state). Shipments with similar origin and destination are integrated into a single one. Totally, there are 4365 combined flows from Europe to the United States and 2911 from the United States to Europe Table 4.

Candidate ports

A total of 12 ports are selected as candidate ports: Rotterdam, Antwerp, Hamburg, Bremerhaven, Le Havre and Felixstowe (Europe); New York, Charleston, Houston, Norfolk, Savannah and Baltimore (USA). These are the main ports on the Trans-Atlantic trade. Connections between hinterlands and candidate ports are presented in Table 5.

Table 4: Hinterland allocation

Country	*Hinterlands*	*Country*	*Hinterlands*
Belgium	181	Norway	45
Denmark	100	Sweden	114
Germany	1019	The United Kingdom	835
France	653	The United States	46
The Netherlands	232	—	

Table 5: Connection between hinterlands and ports

Hinterland	*Port*	*Connection mode*
The United States	US ports	Truck and rail
Continental Europe	Rotterdam, Bremerhaven, Hamburg, Antwerp, Le Havre	Truck and rail
Continental Europe	Felixstowe	Feeder via one of continental ports
The United Kingdom	Rotterdam, Bremerhaven, Hamburg, Antwerp, Le Havre	Feeder via the port of Felixstowe
Norway	Rotterdam, Bremerhaven, Hamburg, Antwerp, Le Havre, Felixstowe	Feeder via the port of Oslo
Denmark	Rotterdam, Bremerhaven, Hamburg, Antwerp, Le Havre, Felixstowe	Feeder via the port of Aarhus
Sweden	Rotterdam, Bremerhaven, Hamburg, Antwerp, Le Havre, Felixstowe	Feeder via the port of Gothenburg

Sea distances (miles): Nautical distances between ports are retrieved from the online tool of Dataloy (www.dataloy.com/).

Inland distances (km): Distances between hinterlands and ports are retrieved from the online tool of Google Map (maps.google.com/). It is assumed that rail and road distances are similar and the inland transport speed is assumed to be 70 kilometres per hour.

Ship costs: Include capital cost, operating cost and fuel cost.

Daily capital cost (USD): $cost = 22.89*(ship_size)^{0.70}$ (Tran and Haasis, 2015a).

Daily operating cost (USD): $cost = 267*(ship_size)^{0.40}$ (Tran and Haasis, 2015a).

Daily fuel consumption at sea (tonne): $sea_fuel = 1.21*(ship_size)^{0.51}$ (Tran and Haasis, 2015a). The price of fuel is assumed at $400 per tonne. In addition to fuel consumption at sea, shipping lines have to cover the consumption in port of auxiliary engine. According to the simulations of Drewry (2009) on East-West routes, fuel cost in port merely plays a marginal part in the total fuel cost (less than 2.7 per cent). In this research, we assume that fuel consumption in port is 5 per cent of the level at sea (adapted from Cariou, 2011).

Port costs

Port dues (USD per ship call): Port dues are retrieved from the database of ENERPLAN, a joint project of Maersk Line, DTU Management Engineering and the IT University of Copenhagen (Brouer *et al*, 2014). Dues include fixed cost per call and variable cost based on ship capacity.

Terminal handling charges (USD per movement): Terminal handling charges are retrieved from the online tool of Hamburg Süd (ecom.hamburgsud.com/ecom/de/ecommerce_portal/tarifs_and_surcharges/thc_calc/ep_thc_calculator.xhtml).

Inland transport costs

(1), (2): adapted from Schade *et al* (2006) and converted from Euro to USD (exchange rate of 1.28). (3): On the basis of the ratios between road transport and rail transport (by tonne-km) from EU Statistics and US Department of Energy (2013). (5): According to UNCTAD (2013), the global volume was 155m TEUs, in correspondence with 1.578m tonnes, so one TEU is roughly equal to 10 tonnes.

Inventory costs

Inventory cost rate: According to Notteboom (2006), cargo delay results in two costs: opportunity cost (3–4 per cent of cargo value per year) and economic

depreciation (10–30 per cent per year). On the basis of this, we assume inventory cost rate of 25 per cent per year.

Container costs: US$0.8 per day per TEU, based on TEU container price of $2890 (Tran and Haasis, 2015a) and assumed economic life of 10 years.

An important component of inventory costs is warehousing cost. In this research, it is assumed that container flows between hinterlands are available, or their origins and destinations are fixed. Therefore, the lack of warehousing cost does not impact on network design, which mainly deals with the selection of ports on shipping routes, as well as the connections between hinterlands and loading and unloading ports.

CO_2 costs

CO_2 tax: $32 per emission tonne (adapted from Schrote *et al* (2011) and converted from Euro to USD).

For ship operations, CO_2 emission is computed by multiplying fuel consumption by a factor of 3.17 (adapted from Psaraftis and Kontovas, 2009).

For cargo operations in port: CO_2 emission is assumed 17.29 kg per TEU (adapted from Veidenheimer, 2011).

For inland transportation: according to Cefic and ECTA (2011), road CO_2 emission is 62 g/tonne-km and rail CO_2 emission is 22 g/tonne-km. On the basis of the ratio of road/rail transportation in Europe and the United States, and the average weight per TEU (Table 6), CO_2 emission is calculated to be 0.54 kg/TEU-km in Europe and 0.45 kg/TEU-km in the United States.

Other operational parameters

Ship speed: 20 knots.

Handling productivity: 180 moves per hour (in all ports, 6 gantry cranes with a productivity of 30 moves per hour each are deployed to serve mainline ships). These figures are based on some works related to handling operation in mainline ports (Baird, 2002; Dragovic *et al*, 2007, 2010; Drewry, 2010). The operation of 6 gantry cranes may not be necessary as the ships on the Trans-Atlantic trade are

Table 6: Inland transport costs

	Road ($/tonne-km) (1)	*Rail ($/tonne-km) (2)*	*Road – Rail ratio (3)*	*Inland transport cost ($/tonne-km) (4)*	*Tonne per TEU (5)*	*Inland transport cost ($/TEU-km) (6)*
EU	0.1792	0.1408	4:1	0.174	10	1.74
The United States	0.256	0.0128	4:3	0.155	10	1.55

often smaller than those on the Trans-Pacific and Europe-Asia trades. Nevertheless, to simulate the variation of cost and time in ports as ship capacity becomes bigger (from 2000 TEUs to 18 000 TEUs), we assume such operational condition for all cases.

Ship utilisation on the main leg: 80 per cent.

Manoeuvring time in ports: 3 hours per entry/exit (adapted from Wijnolst *et al*, 2000).

Export/import dwell time in ports: 3 days (adapted from Rankine, 2003).

Feeder transportation: a feeder ship (1000 TEUs) connects directly feeder and transhipment ports; time in port is assumed to be 1 day in each call.

Computational Results and Analyses

Results

Our model is run with various ship sizes from 2000 TEUs to 18 000 TEUs. According to the database of Drewry (2015), the average ship size of

Table 7: Summary of computational experiments

Size (TEU)	*Route*	*Time (day)*		*Unit cost (USD per TEU)*						*Error (%)*
		At sea	*In port*	*Ship costs*	*Port costs*	*Inventory costs*	*CO_2 costs*	*Inland/feeder transport costs*	*Total cost*	
2000	(1)	25.39	3.75	327	520	694	66	917	2523	3.2
3000	(1)	25.39	4.13	270	504	705	56	917	2451	3.5
4000	(1)	25.39	4.51	237	495	715	50	917	2414	3.8
5000	(2)	25.47	4.88	215	491	723	46	919	2393	4.0
6000	(2)	25.47	5.26	198	488	733	43	919	2381	4.3
7000	(2)	25.47	5.63	185	485	744	40	919	2373	4.5
8000	(2)	25.47	6.01	174	484	754	39	919	2370	4.8
9000	(2)	25.47	6.39	166	482	765	37	919	2369	5.1
10 000	(2)	25.47	6.76	159	481	775	36	919	2370	5.4
11 000	(2)	25.47	7.14	152	480	786	35	919	2372	5.6
12 000	(2)	25.47	7.52	147	480	796	34	919	2375	5.9
13 000	(2)	25.47	7.89	142	479	807	33	919	2380	6.2
14 000	(2)	25.47	8.27	138	478	817	32	919	2385	6.4
15 000	(2)	25.47	8.65	135	478	828	31	919	2390	6.7
16 000	(2)	25.47	9.02	131	477	838	31	919	2397	7
17 000	(2)	25.47	9.40	128	477	849	30	919	2403	7.2
18 000	(2)	25.47	9.77	125	477	859	29	919	2410	7.5

Note: Route (1): Antwerp→ Bremerhaven→Hamburg→Rotterdam→Felixstowe→Le Havre→New York→ Norfolk→Charleston→Houston→Savannah→Baltimore→Antwerp (12 189 miles)
Route (2): Antwerp→Felixstowe→Bremerhaven→Hamburg→Rotterdam→Le Havre→New York→Norfolk→ Charleston→Houston→Savannah→Baltimore→Antwerp (12 225 miles)

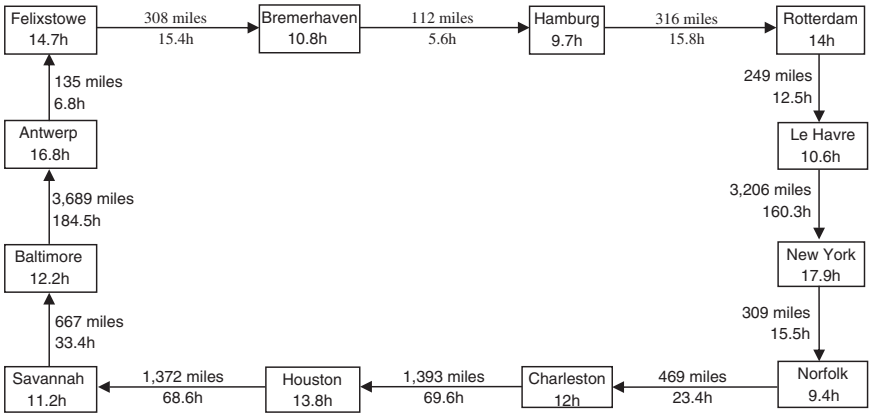


Figure 2: The optimal voyage of a 9000 TEU ship.

Table 8: Cargo flows through ports

Port	*Loading*				*Unloading*			
	Shipments	*TEUs*	*Boxes*	*Value ($)*	*Shipments*	*TEUs*	*Boxes*	*Value ($)*
Antwerp	849	1460	869	57 063 018	919	1857	1068	110 177 163
Felixstowe	1455	1223	755	65 151 558	1455	1457	812	67 238 511
Bremerhaven	110	885	559	57 870 343	88	563	313	35 844 783
Hamburg	428	872	504	50 743 249	132	281	168	14 355 064
Rotterdam	714	1670	996	97 195 814	135	783	449	32 422 156
Le Havre	809	1088	667	70 426 954	182	276	155	12 108 330
New York	285	652	374	31 444 503	1470	2908	1766	163 961 954
Norfolk	152	76	42	1 403 393	547	962	567	58 462 269
Charleston	103	127	66	2 230 505	996	1679	1022	109 120 142
Houston	664	1219	725	60 402 341	806	1098	687	50 482 041
Savannah	531	1419	800	64 400 271	222	250	141	7 951 256
Baltimore	1176	1725	958	112 264 994	324	302	165	8 473 273

Trans-Atlantic routes falls between 2000 and 9000 TEUs. In addition to this range, our article also considers the potential of bigger ship sizes on the trade.

The solutions are found by specifically tailored computer programs written in the Freepascal language. Some key outcomes of the experiment are presented in Table 7. The lowest total cost per TEU falls in the case of a 9000 TEU ship ($2369), whereas the most expensive one is the option of a 2000 TEU ship ($2523). Ships operate optimally in two different routes embracing 12 ports of call. Of the five main cost components, inland/feeder transport costs account for the biggest part with more than 35 per cent of the total cost. CO_2 costs contribute the smallest, with less than 2.5 per cent of total costs. Inventory costs and shipping costs (ship costs and port costs) each account for around one third of

Table 9: Breakdown of total unit cost (9000 TEU ship)

Cost item	*Sub-component*	*Value ($ per TEU)*	*Percentage*
Ship costs	Capital cost at sea	27.51	1.16
	Operating cost at sea	20.9	0.88
	Bunker cost at sea	103.99	4.39
	Capital cost in port	6.9	0.29
	Operating cost in port	5.24	0.22
	Bunker cost in port	1.3	0.05
Port costs	Port due and terminal handling charge	482.31	20.36
Inventory costs	Inventory cost at sea	371.91	15.7
	Inventory cost in port (during ship operation)	106.94	4.51
	Inventory cost in port (during dwell time)	226.77	9.57
	Inventory cost during inland transport	22.03	0.93
	Safety stock	36.91	1.56
Inland/feeder transport costs	Inland/feeder transport	919.17	38.8
CO_2 costs	CO_2 cost at sea	26.37	1.11
	CO2 cost in port (by ship operation)	0.33	0.01
	CO2 cost in port (by cargo operation)	1.11	0.05
	CO2 cost during inland transport	9.17	0.39

the total cost. The model error is calculated by the ratio between the bypassed costs in the greedy algorithm (ship cost, CO_2 cost and inventory cost in port) and total cost. The indicator is less than 7.5 per cent in all cases.

The use of a 9000 TEU ship results in the smallest total unit cost. Figure 2 illustrates its voyage, involving 25.47 days at sea and 6.39 days in ports. Table 8 shows the breakdown of the total unit cost based on cost item and supply chain stage. Table 9 demonstrates cargo flows going through the visited ports on the route. Import containers tend to be unloaded more in the first called regional ports (Antwerp and Felixstowe in Europe, New York in the United States), whereas export containers tend to be loaded in the last ones (Rotterdam, Le Havre in Europe; Savannah and Baltimore in the United States). To decrease inventory cost, it is reasonable to unload containers as soon as possible and load them as late as possible. Studies of Lago *et al* (2001) and Malchow (2001) for the US ports also indicate that a port could attract more export cargo when it is called last, and more import cargo when it is called first, in a shipping service.

Analyses

Ship costs and port costs are the most conventional factors in network optimisation problems. In fact, they only represent less than one third of the door-to-door cost. The deployment of mega vessels may drive down ship costs and port costs

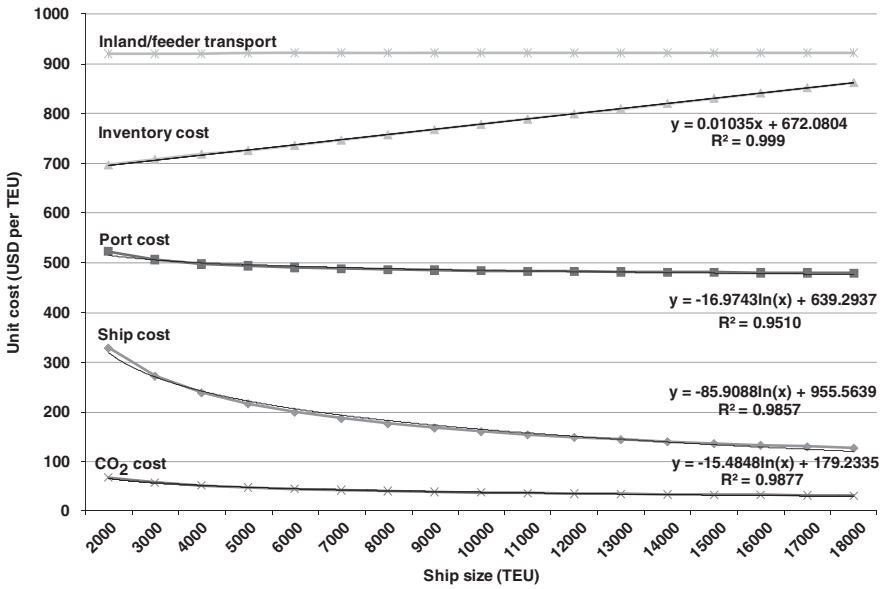


Figure 3: Ship size and unit costs.

per TEU. According to our model, doubling vessel capacity will lower them by \$59.55 and \$11.77, respectively Figure 3.[2]

In spite of their positive effects on shipping costs, mega vessels lead to higher inventory costs. If ship size goes up by 1000 TEUs, inventory costs per TEU increase by \$10.35. Inventory costs are a hidden factor and often out of consideration in optimisation problems. Nevertheless, they accounts for a significant portion of total cost and increases with ship size (27.5 per cent with a 2000 TEU vessel, 32.2 per cent with 10 000 TEU and 35.7 per cent with 18 000 TEU). Optimal ship size is evidently a trade-offs between shipping and inventory costs.

The largest inventory costs are incurred during transit time at sea and dwell time in ports. The former is determined by ship speed whereas the latter by port policy and productivity. However, the major disadvantages of bigger vessels come from inventory costs generated during ship operations in port and safety stock. On the one hand, bigger vessels require longer operating time in ports. On the other hand, they make longer the interval between two consecutive trips to ports, which means that shippers must keep higher stock to prevent the depletion of inventory. Total port time is 6.4 days for a 9000 TEU ship, in comparison with 9.8 days for an 18 000 TEU ship. Consequently, unit inventory cost in port time increases from \$107 to \$165. Service frequency is 2 days for the

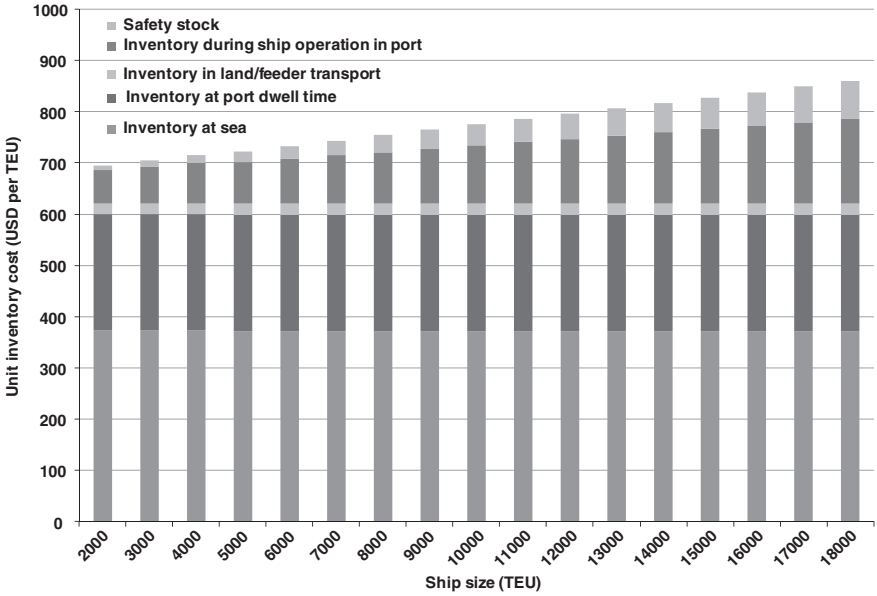


Figure 4: Ship size and unit inventory cost.

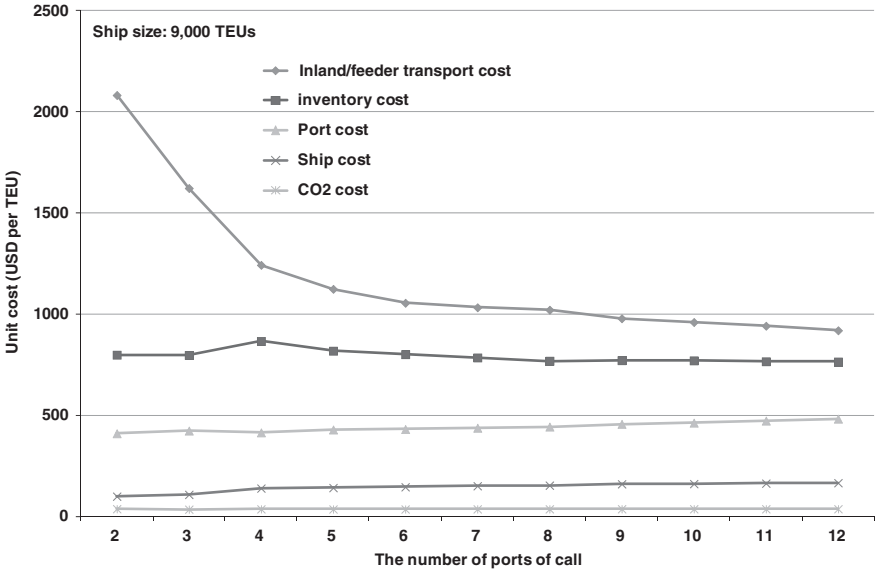


Figure 5: The number of ports of call and unit cost.

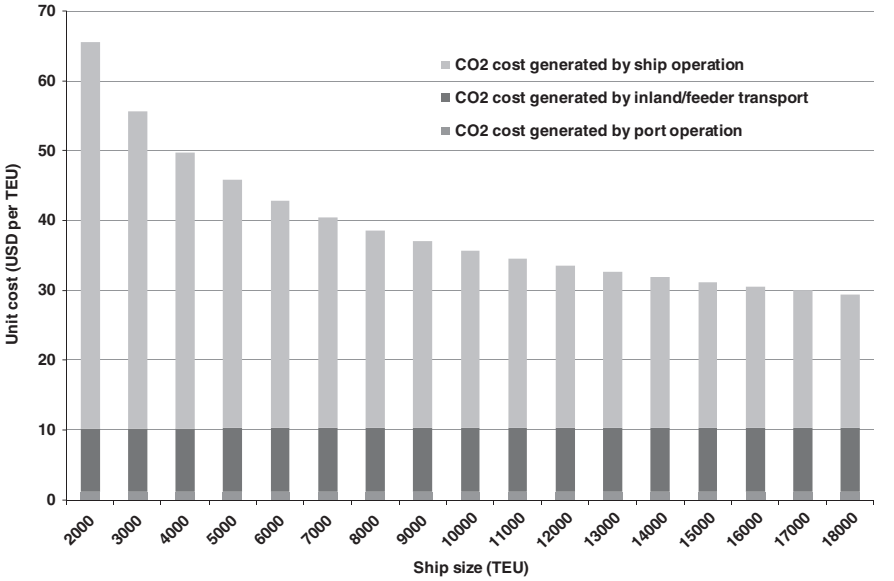


Figure 6: Ship size and unit CO_2 cost.

former option and 4 days for the latter, which is equivalent to safety stock costs of $36.91 and $73.82 per TEU, correspondingly Figure 4.

Although inland/feeder time plays only a small part in total transit time, inland/feeder transport costs are the most expensive item in the door-to-door cost, with a share of over 35 per cent. These costs are naturally influenced by the choice of ports on shipping routes; the more the port calls, the closer the service to the final consumer. Consequently, transport costs between hinterlands and visited ports declines. As illustrated in Figure 5, unit inland/feeder transport costs are $2082 on the route with 2 port calls, while the respective costs on the routes with 7 and 12 port calls are $1033 and $919. High distribution costs are obviously a weakness in routes with a few visits, although these routes benefit from lower shipping costs because of shorter travelling distances, as well as the smaller number of visited ports.

The environmental effects of shipping have become a growing concern in recent years (Cariou, 2011; Eide *et al*, 2011; Song and Xu, 2012; Balland *et al*, 2014; Wo and Moon, 2014). Our model takes account of CO_2 costs through bunker consumption and CO_2 tax. Actually, they contribute merely a marginal share in total costs with 2.6 per cent ($65.57) in the case of a 2000 TEU ship and 1.2 per cent ($29.42) of an 18 000 TEU ship. It can be realized that unit environment costs go down as ship capacity increases. The largest part of CO_2

costs in the supply chain is generated by ship operations. Therefore, lower environmental costs could be a good argument to launch mega ships. It has been claimed by Maersk Line that CO_2 emission of its Triple-E ships is 50 per cent less than the industry average on the Asia–Europe trade lane Figure 6.

Conclusions

Route design in container shipping is a challenging task because of the many constraints, as well as the conflicting goals of different stakeholders. This article has proposed a model to optimise transportation flows between two regions, taking into account not only shipping and port costs, but also those of shippers (inland/feeder transport costs, time costs of cargo) and society (CO_2 costs). In terms of supply chain stage, the model integrates both maritime and inland legs. The research was carried out using trade flows between the United States and Europe. The optimal solution corresponds to the case of 12 ports on the shipping route and a ship capacity of 9000 TEUs.

The computational outcome reveals that inland/feeder transport costs are the highest among supply chain costs and they are influenced much by the selection of ports. The greater the number of port calls, the lower these costs are. The high effect of inland/feeder transport costs implies an advantage of ports close to final markets in attracting shipping calls, as well as that of the route pattern with multi-port calls.

Ship costs, port dues and terminal handling charges are major burdens of liner operators in terms of organising shipping services. Nevertheless, they are only a part of the whole supply chain game. Lowering these costs does not necessarily mean that total costs will become smaller. There is little about the advantage of mega vessels in decreasing maritime costs. In addition, their deployment also results in higher inventory costs, and possibly leads to higher total costs.

Although often treated as hidden costs, inventory costs represent a considerable share of total costs. The level of these costs is a major barrier for the deployment of mega vessels as a result of longer operations in ports, which increase inventory costs during port time, and longer waiting times between two successive trips, resulting in higher safety stock. Optimal vessel capacity is mainly a balance between inventory and shipping costs.

Our research emphasises that it is necessary to include not only shipping factors, but also other ones in determining optimal transport systems. To design shipping services, liner carriers must consider not only their own costs, but also their customers' costs, and even society costs. Otherwise, their optimal services in maritime chains may become non-optimal with the overall supply chain.

Acknowledgements

The authors are thankful to the editors and reviewers of *MEL* for many detailed comments and corrections; they have been extremely helpful in terms of improving the quality of this article. The authors are also thankful to Mr Tonny Bennet (PIERS – UK) for providing container trade data between Europe and the United States; this research could not have been done without it. Many thanks are also due to numerous experts for their advice regarding the more practical aspects of shipping operations. The co-operative junior research group on Computational Logistics is funded by the University of Bremen in line with the Excellence Initiative of German federal and state governments.

Notes

1 Our model would obviously have been stronger if ship size were a decision variable. However, without the assumption of fixed size, the solution space would become very large and it is infeasible to find a good solution.
2 On the one hand, mega vessels result in scale economies at sea. On the other hand, they cause diseconomies in ports. Whereas the saving of ship costs has been widely accepted, that of port costs has been still questionable, especially with high pressure of mega vessels on port investment and operation (see more in Dragovic *et al*, 2010; Saanen, 2013; Tran and Haasis, 2015a). In our research, port costs are mainly retrieved from Brouer *et al* (2014). According to their work, unit port costs will decrease as ships become bigger. This outcome may be debatable. Nevertheless, as far as we know, their model is the only one in the market that can provide cost estimation of key ports in the world. In addition, it is created using operational data of Maersk Line, which more or less ensures its reasonableness.

References

Agarwal, R. and Ergun, O. (2008) Ship scheduling and network design for cargo routing in liner shipping. *Transportation Science* 42(2): 175–196.

Agarwal, R. and Ergun, O. (2010) Network design and allocation mechanisms for carrier alliances in liner shipping. *Operations Research* 58(6): 1726–1742.

Alvarez, J.F. (2009) Joint routing and deployment of a fleet of container vessels. *Maritime Economics & Logistics* 11: 186–208.

Aversa, R., Botter, R.C., Haralambides, H.E. and Yoshizaki, H.T.Y. (2005) A mixed integer programming model on the location of a hub port in the East Coast of South America. *Maritime Economics & Logistics* 7: 1–18.

Balland, O., Erikstad, S.O. and Fagerholt, K. (2014) Optimized selection of air emission controls for vessels. *Maritime Policy & Management* 39(4): 387–400.

Baird, A.J. (2002) The economics of transhipment in Northern Europe. *International Journal of Maritime Economics* 4: 249–280.

Baird, A.J. (2006) Optimising the container transhipment hub location in northern Europe. *Journal of Transport Geography* 14(3): 195–214.

Bell, M.G.H., Liu, X., Angeloudis, P., Fonzone, A. and Hosseinloo, S.H. (2011) A frequency-based maritime container assignment model. *Transportation Research Part B* 45(8): 1152–1161.

Bell, M.G.H., Liu, X., Rioult, J. and Angeloudis, P. (2013) A cost-based maritime container assignment model. *Transportation Research Part B* 58: 58–70.

Brouer, B.D., Alvarez, J.F., Plum, C.E.M., Pisinger, D. and Sigurd, M.M. (2014) A base integer programming model and benchmark suite for liner shipping network design. *Transportation Science* 48(2): 281–312.

Brouer, B.D., Dirksen, J., Pisinger, D., Plum, C.E.M. and Vaaben, B. (2012) The vessel schedule recovery problem (VSRP) – A MIP model for handling disruptions in liner shipping. *European Journal of Operational Research* 224(2): 362–374.

Carbone, V. and De Martino, M. (2003) The changing role of ports in supply-chain management: An empirical analysis. *Maritime Policy & Management* 30(4): 305–320.

Cariou, P. (2011) Is slow steaming a sustainable means of reducing CO2 emissions from container shipping? *Transportation Research Part D* 16(3): 260–264.

Cariou, P. and Cheaitou, A. (2014) Cascading effects, network configurations, and optimal transshipment volumes in liner shipping. *Maritime Economics and Logistics* 16(3): 321–342.

Cefic and ECTA. (2011) Guidelines for measuring and managing CO_2 emission from freight transport operations, http://www.cefic.org/Documents/IndustrySupport/Transport-and-Logistics/Best%20Practice%20Guidelines%20-%20General%20Guidelines/Cefic-ECTA%20Guidelines%20for%20measuring%20and%20managing%20CO2%20emissions%20from%20transport%20operations%20Final%2030.03.2011.pdf, accessed 19 August 2014.

Christiansen, M., Fagerholt, K. and Ronen, D. (2004) Ship routing and scheduling: Status and perspectives. *Transportation Science* 38(1): 1–18.

Christiansen, M., Fagerholt, K., Nygreen, B. and Ronen, D. (2007) Maritime transportation. In: C. Barnhart and G. Laporte (eds.) *Invited chapter to Handbooks in Operations Research and Management Science*. Transportation, Amsterdam: North Holland, Vol. 14, pp. 189–284.

Christiansen, M., Fagerholt, K., Nygreen, B. and Ronen, D. (2013) Ship routing and scheduling in the new millennium. *European Journal of Operational Research* 228(3): 467–483.

Cullinane, K. and Wang, Y. (2012) The hierarchical configuration of the container port industry: An application of multiple linkage analysis. *Maritime Policy & Management* 39(2): 169–187.

Dragovic, B., Ryoo, D.-K., Park, N.-K. and Radmilovic, Z. (2007) Container ship development: A review of state of the art. In Proceedings of IAME conference, Athen, Greece.

Dragovic, B., Zrnic, B. and Ryo, D.K. (2010) An overview of relationship between container ships and quay cranes in port. In Proceedings of IAME conference, Lisbon, Portugal.

Drewry. (2009) *Capacity Management – Surviving the Container Crisis*. London: Drewry Shipping Consultants.

Drewry. (2010) *Container Terminal Capacity and Performance Benchmarks*. London: Drewry Shipping Consultants.

Drewry. (2015) *Container Forecaster*. London: Drewry Shipping Consultants.

Ducruet, C. (2013) Network diversity and maritime flows. *Journal of Transport Geography* 30: 77–88.

Eide, M.S., Longva, T., Hoffmann, P., Endresen, O. and Dalsoren, S.B. (2011) Future cost scenarios for reduction of ship CO2 emissions. *Maritime Policy & Management* 38(1): 11–37.

Ferrari, C., Parola, F. and Benacchio, M. (2008) Network economies in liner shipping: The role of home markets. *Maritime Policy & Management* 35(2): 127–143.

Fremont, A. (2007) Global maritime networks – The case of Maersk. *Journal of Transport Geography* 15: 431–442.

Gelareh, S. and Meng, Q. (2010) A novel modeling approach for the fleet deployment problem within a short-term planning horizon. *Transportation Research Part E* 46(1): 76–89.

Gelareh, S. and Pisinger, D. (2011) Fleet deployment, network design and hub location of liner shipping companies. *Transportation Research Part E* 47(6): 947–964.

Gelareh, S., Maculan, N., Mahey, P. and Monemi, R.N. (2013) Hub-and-spoke network design and fleet deployment for string planning of liner shipping. *Applied Mathematical Modelling* 37(5): 3307–3321.

Gelareh, S., Nickel, S. and Pisinger, D. (2010) Liner shipping hub network design in a competitive environment. *Transportation Research Part E* 46(6): 991–1004.

Imai, A., Nishimura, E., Papadimitriou, S. and Liu, M. (2006) The economic viability of container mega ships. *Transportation Research Part E* 42(1): 21–41.

Jansson, J.O. and Shneerson, D. (1987) *Liner Shipping Economics*. London and New York: Chapman and Hall.

Kjeldsen, K.H. (2011) Classification of ship routing and scheduling problems in liner shipping. *INFOR* 49(2): 139–152.

Lago, A., Malchow, M. and Kanafani, A. (2001) In Proceedings of the IAME Annual Conference 2001. Hongkong Polytechnic University: Hongkong, China.

Liu, Z., Meng, Q., Wang, S. and Sun, Z. (2014) Global intermodal liner shipping network design. *Transportation Research Part E* 61: 28–39.

Lofstedt, B., Pisinger, D. and Spoorendonk, S. (2010) Liner shipping cargo allocation with repositioning of empty containers. *INFOR* 49(2/5): 109–124.

Malchow, M.B. (2001) An analysis of port selection. Unpublished Phd's thesis, University of California, Berkeley, CA.

Meng, Q. and Wang, S. (2011a) Optimal operating strategy for a long-haul liner service route. *European Journal of Operational Research* 215(1): 105–114.

Meng, Q. and Wang, S. (2011b) Liner shipping service network design with empty container repositioning. *Transportation Research Part E* 47(5): 695–708.

Meng, Q. and Wang, T. (2011c) A scenario-based dynamic programming model for multi-period liner ship fleet planning. *Transportation Research Part E* 47(4): 401–413.

Meng, Q., Wang, S., Andersson, H. and Thun, K. (2013) Containership routing and scheduling in liner shipping: Overview and future research directions. *Transportation Science* 48(2): 265–280.

Meng, Q., Wang, T. and Wang, S. (2015) Multi-period liner ship fleet planning with dependent uncertain container shipment demand. *Maritime Policy & Management* 42(1): 43–67.

Mourao, M.C., Pato, M.V. and Paixao, A.C. (2002) Ship assignment with hub and spoke constraints. *Maritime policy & Management* 29(2): 135–150.

Mulder, J. and Dekker, R. (2014) Methods for strategic liner shipping network design. *European Journal of Operational Research* 235(2): 367–377.

Notteboom, T.E. (2006) The time factor in liner shipping services. *Maritime Economics & Logistics* 8: 19–39.

Notteboom, T.E. and Rodrique, J.P. (2008) Containerisation, box logistics and global supply chains: The integration of ports and liner shipping networks. *Maritime Economics & Logistics* 10: 152–174.

Pettit, S.J. and Beresford, A.K.C. (2009) Port development: From gateways to logistics hubs. *Maritime Policy & Management* 36(3): 253–267.

Plum, C.E.M., Pisinger, D. and Sigurd, M.M. (2014) A service flow model for the liner shipping network design problem. *European Journal of Operational Research* 235(2): 378–386.

Polat, O., Guenther, H.O. and Kulak, O. (2014) The feeder network design problem: Application to container services in the Black Sea region. *Maritime Economics & Logistics* 16(3): 343–369.

Psaraftis, H.N. and Kontovas, C.A. (2009) CO2 emission statistics for the world commercial fleet. *WMU Journal of Maritime Affairs* 8(1): 1–25.

Qi, X. and Song, D.P. (2012) Minimizing fuel emissions by optimizing vessel schedules in liner shipping with uncertain port times. *Transportation Research Part E* 48(4): 863–880.

Rankine, G. (2003) Benchmarking container terminal performance. In Proceedings of Container Port Conference, Rotterdam, The Netherlands.

Reinhardt, L.B. and Pisinger, D. (2012) A branch and cut algorithm for the container shipping network design problem. *Flexible Services and Manufacturing Journal* 24(3): 349–374.

Reinhardt, L.B., Kallehauge, B., Norrelund, A. and Olsen, A. (2007) Network design models for container shipping, http://orbit.dtu.dk/en/publications/network-design-models-for-container-shipping%2848c51176-5e2a-4247-bdca-d78f56cbcc05%29.html, accessed 19 August 2011.

Robinson, R. (1998) Asian hub/feeder nets: the dynamics of restructuring. *Maritime Policy & Management* 25(1): 21–40.

Robinson, R. (2002) Ports as elements in value-driven chain systems: The new paradigm. *Maritime Policy &Management* 29(3): 241–255.

Robinson, R. (2006) Port-oriented landside logistics in Australian ports: A strategic framework. *Maritime Economics & Logistics* 8: 40–59.

Saanen, I. (2013) Mega ships: Positive asset or terminals' worst nightmare?, http://www.porttechnology.org/technical_papers/mega_ships_positive_asset_or_terminals_worst_nightmare, accessed 6 August 2013.

Schade, W. *et al* (2006) COMPETE Final Report: Analysis of the contribution of transport policies to the competitiveness of the EU economy and comparison with the United States, http://ec.europa.eu/ten/transport/studies/doc/compete/compete_report_en.pdf, accessed 19 August 2014.

Schroten, A., Essen, H.V. and Anthes, R. (2011) External cost calculator – Methodology report. Delft: CE Delft.

Seoane, M.J.F., Laxe, F.G. and Montes, C.P. (2013) Foreland determination for containership and general cargo ports in Europe (2007–2011). *Journal of Transport Geography* 30: 56–67.

Shintani, K., Imai, A., Nishimura, E. and Papadimitriou, S. (2007) The container shipping network design problem with empty container repositioning. *Transportation Research Part E* 43(1): 39–59.

Song, D.P. and Dong, J.X. (2012) Cargo routing and empty container repositioning in multiple shipping service routes. *Transportation Research Part B* 46(10): 1556–1575.

Song, D.P. and Dong, J.X. (2013) Long-haul liner service route design with ship deploymentand empty container repositioning. *Transportation Research Part B* 55: 188–211.

Song, D.P. and Xu, J. (2012) An operational activity-based method to estimate CO2 emissions from container shipping considering empty container repositioning. *Transportation Research Part D* 17(1): 91–96.

Tran, N.K. (2011) Studying port selection on liner routes: An approach from logistics perspective. *Research in Transportation Economics* 32(1): 39–53.

Tran, N.K. and Haasis, H.D. (2014) Empirical analysis of the container liner shipping network on the East-West corridor (1995–2011). *Netnomics* 15(3): 121–153.

Tran, N.K. and Haasis, H.D. (2015a) An empirical study of fleet expansion and growth of ship size in container liner shipping. *International Journal of Production Economics* 158: 241–253.

Tran, N.K. and Haasis, H.D. (2015b) Literature survey of network optimization in container liner shipping. *Flexible Services and Manufacturing Journal* 27(2): 139–179.

UNCTAD. (1992) *Port Marketing and the Challenge of the Third Generation Port*. Geneva, Switzerland: UNCTAD.

UNCTAD. (2013) *Review of Maritime Transport*. Geneva, Switzerland: United Nations.

US Department of Energy. (2013) Freight Transportation Modal Shares: Scenarios for a Low-Carbon Future, http://www.ourenergypolicy.org/freight-transportation-modal-shares-scenarios-for-a-low-carbon-future/, accessed 19 August 2014.

Veidenheimer, K. (2011) *Carbon Dioxide Emission in Maritime Transport*. Hamburg, Germany: Anchor Academic Publishing.

Verny, J. and Grigentin, C. (2009) Container shipping on the Northern Sea Route. *International Journal of Production Economics* 122: 107–117.

Wang, B. and Tang, G. (2010) Stochastic optimization model for container shipping of sea carriage. *Journal of Transportation Systems Engineering and Information Technology* 10(3): 58–63.

Wang, H., Wang, S. and Meng, Q. (2014) Simultaneous optimization of schedule coordination and cargo allocation for liner container shipping networks. *Transportation Research Part E* 70: 261–273.

Wang, S. and Meng, Q. (2012c) Liner ship route schedule design with sea contingency time and port time uncertainty. *Transportation Research Part B* 46(5): 615–633.

Wang, S. and Meng, Q. (2014) Liner shipping network design with deadlines. *Computers & Operations Research* 41: 140–149.

Wang, S. (2013) Essential elements in tactical planning models for container liner shipping. *Transportation Research Part B* 54: 84–99.

Wang, S. (2014) A novel hybrid-link-based container routing model. *Transportation Research Part E* 61: 165–175.

Wang, S., Alharbi, A. and Davy, P. (2014) Liner ship route design with port time windows. *Transportation Research Part C* 41: 1–17.

Wang, S., Liu, Z. and Bell, M.G.H. (2015) Profit-based maritime container assignment models for liner shipping networks. *Transportation Research Part B* 46(5): 59–76.

Wijnolst, N., Waals, F., Bello, F., Gendroneau, Y. and Kempen, D.V. (2000) *Malacca-max [2] Container Shipping Network Economy*. Delft, The Netherlands: DUP.

Woo, J.K. and Moon, D.S.H. (2014) The effects of slow steaming on the environmental performance in liner shipping. *Maritime Policy & Management* 41(2): 176–191.

Zacharioudakis, P.G., Iordanis, S., Lyridis, D.V. and Psaraftis, H.N. (2011) Liner shipping cycle cost modelling, fleet deployment optimization and what-if analysis. *Maritime Economics & Logistics* 13: 278–297.

Yang, Z., Chen, K. and Notteboom, T. (2012) Optimal design of container liner services: Interactions with the transport demand in ports. *Maritime Economics & Logistics* 14: 409–434.